

**ArcDoc: Angular Margin Optimization
for Zero-Shot Document Classification**
Seminar — Updated Results



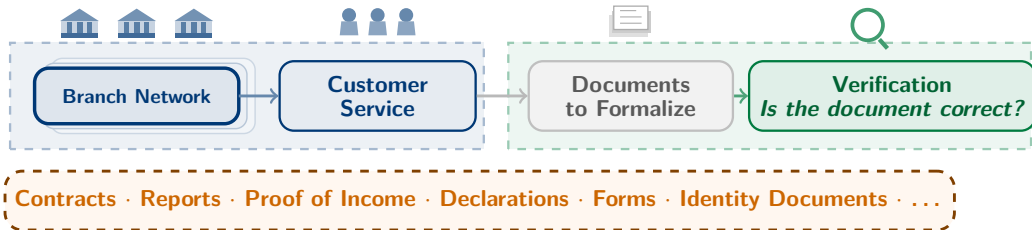
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Dataset Structure

- **144** unique classes · **4 993** documents
- **5** cross-validation splits (S0–S4) + **1** reserved split (S5)
- **120** training classes / **24** *novel* classes per split
- Test classes **disjoint** from training classes

Formalization

Learn $f_\theta : \mathcal{X} \rightarrow \mathcal{S}^{d-1}$ such that, for any $x, x' \in \mathcal{X}$:

$\langle f_\theta(x), f_\theta(x') \rangle$ high if same class, low otherwise

- including for classes in $\mathcal{C}_{\text{test}}$ **never seen** during training
- structurally **identical** to open-set face recognition

Zero-Shot Protocol

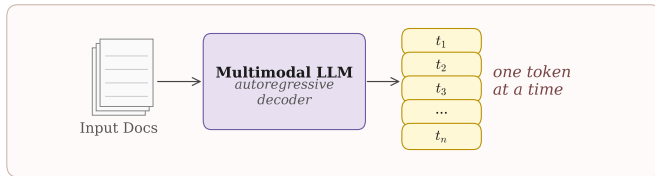
$$\mathcal{C}_{\text{train}} \cap \mathcal{C}_{\text{test}} = \emptyset$$

Metric: **EER** (Equal Error Rate) on pairwise verification.
Lower is better.

Macedo, Costa et al. *Visual Document Matching for Zero-Shot Document Classification*. ICDAR 2025 Workshops, LNCS 16225, pp. 192–208. DOI: 10.1007/978-3-032-09368-4_12



(A) Generative Model — Autoregressive Decoding



(B) ArcDoc — At-Once Aggregation

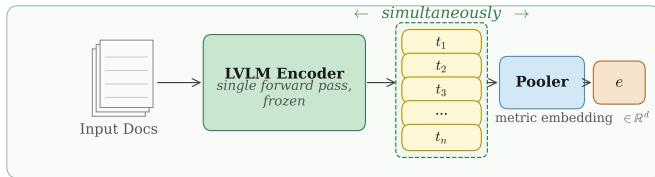


Figure: At-Once Aggregation vs. Autoregressive Generation. (A) A standard LVLM generates tokens one by one — comparing documents requires *post-hoc* text matching on the generated output. (B) **ArcDoc** extracts all representations in a single *forward pass* and aggregates them into a compact embedding; similarity is computed directly in the metric space, with no generative decoding.

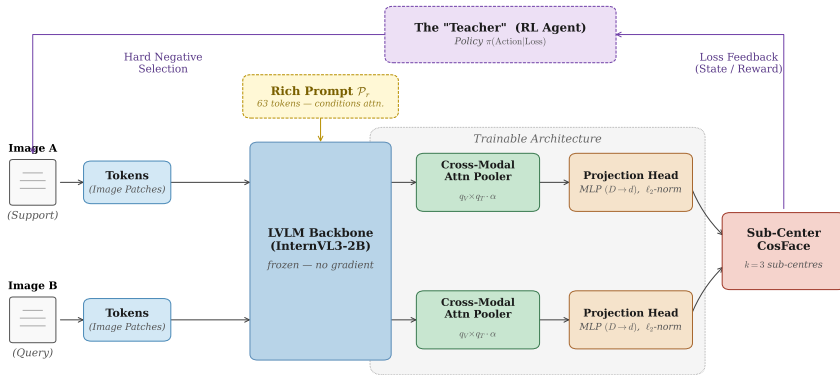


Figure: Frozen LVLM backbone (InternVL3-2B), conditioned on a rich descriptive prompt $\mathcal{P}_r \rightarrow$ pooling head (Cross-Modal or Mean/MQAP) $\rightarrow$ Residual Projection $\rightarrow$ Sub-Center CosFace (angular margin). An RL Teacher agent actively mines *hard negatives* in a curriculum phase.



Even **before any training**, the frozen backbone's output layer already encodes class-discriminative spatial structure:

- Letters concentrate activation at the heading
- Invoices spread activation across the tabular body
- This pattern is **prompt-invariant** — driven purely by visual content

Quantitatively, mean-pooled cosine similarity improves from **7.62% EER** (input layer) to **4.78%** (output layer) from backbone contextualization alone. **But** visually similar classes remain entangled without an explicit separation objective → this motivates the **metric learning loss** (next).

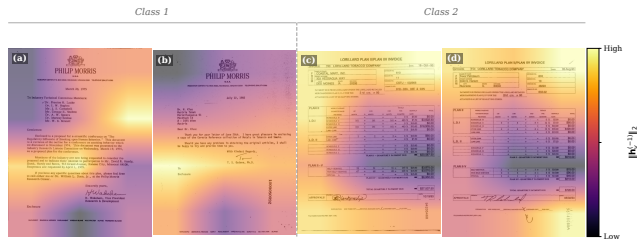
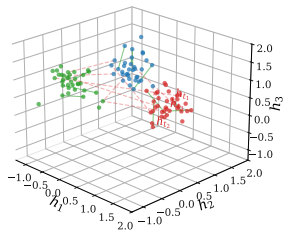


Figure: Visual activation per spatial patch at the backbone's output layer (base model, no metric learning), for two example documents. Letters (left) vs. invoices (right) show distinct, class-specific spatial signatures — this is the representation the loss functions (next) are trained to sharpen.

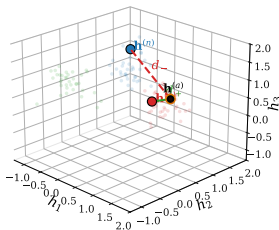


(a) Contrastive Loss



$$\mathcal{L} = \frac{1}{2} [y d^2 + (1 - y) \max(m - d, 0)^2]$$

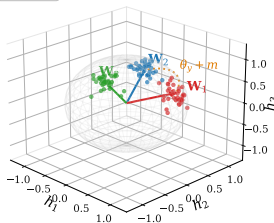
(b) Triplet Loss



$$\mathcal{L} = \max(d_+ - d_- + m, 0)$$

(c) ArcFace

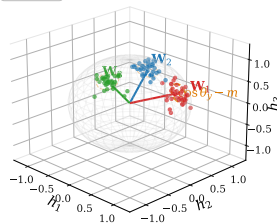
$$\|h_t\| = 1$$



$$\mathcal{L} = -\log \frac{e^{\cos(\theta_y + m)}}{e^{\cos(\theta_y + m)} + \sum_{j \neq y} e^{\cos(\theta_j)}}$$

(d) CosFace

$$\|h_t\| = 1$$



$$\mathcal{L} = -\log \frac{e^{s(\cos(\theta_y) - m)}}{e^{s(\cos(\theta_y) - m)} + \sum_{j \neq y} e^{s(\cos(\theta_j) - m)}}$$

- Token embeddings h_t — Class A ■ Token embeddings h_t — Class C --- Negative pair / d_- (different class) --- Cosine margin $\cos \theta_y - m$ (CosFace)
- Token embeddings h_t — Class B — Positive pair / d_+ (same class) --- Angular margin $\theta_y + m$ (ArcFace) w Class prototypes W_j (unit sphere)

Contrastive/Triplet: no global geometric anchor, clusters form at arbitrary positions and scales. **ArcFace/CosFace:** hypersphere with angular margin — class-agnostic geometry, transferable to unseen classes.

Both ArcFace and CosFace are extended with **Sub-Center** ($k=3$ sub-prototypes per class) to handle multi-modal intra-class variance. *Circle Loss* was also evaluated but eliminated (high variance in fine search).



Three Phases (Bayesian Sweeps)

1. **Coarse** — 173 runs; two-gate filter (5% EER ceiling + divergence check); **68.8%** pass rate
2. **Fine** — 96 runs across 10 groups; LR range narrowed to $2\text{--}10 \times 10^{-5}$
3. **Full Evaluation** — champion config trained with full epochs across all 5 CV splits

Key Findings

- **LR** is the dominant hyperparameter across all groups ($|r| > 0.87$ with $\log(\text{lr})$)
- Margin: Sub-Center CosFace $k=3$ converges at $m \approx 0.18\text{--}0.35$; ArcFace $k=3$ best at $m \approx 0.31$
- Scale $s = 32$ stable across families
- *Circle Loss* did not advance (high variance in fine search)

AdamW, ReduceLR0nPlateau. Single GPU:
RTX 4060 Ti (16 GB) / RTX 4090.



	Split 0	Split 1	Split 2	Split 3	Split 4	Mean	
Sub-Center CosFace	1.35 <i>P1</i>	1.33 <i>P2+M</i>	0.33 <i>P2+M</i>	1.52 <i>P2+M</i>	0.21 <i>P2+M</i>	0.95	worst rank (per split) best
Sub-Center ArcFace	1.16 <i>P1</i>	1.85 <i>P1</i>	0.42 <i>P1</i>	1.90 <i>P2+M</i>	0.41 <i>P2+M</i>	1.15	
Triplet	2.22 <i>P1</i>	1.54 <i>P2</i>	0.42 <i>P1</i>	2.67 <i>P1</i>	0.57 <i>P2</i>	1.48	
Contrastive	3.28 <i>P1</i>	1.69 <i>P1</i>	1.71 <i>P1</i>	1.62 <i>P2+M</i>	1.96 <i>P1</i>	2.05	

P1 = best in Phase 1 (epochs 1-10) *P2+M* / *P2* = best in Phase 2 (epochs 11-15), w/ mining / w/o mining lower is better

Figure: Cumulative best EER (across 15 epochs, Phases 1+2) per loss and split on LA-CDIP. **Sub-Center CosFace** achieves the best mean with hard-negative mining (0.95%).

Loss	With Mining	Without Mining	Δ (pp)
Sub-Center CosFace	0.95%	1.22%	+0.27
Sub-Center ArcFace	1.15%	1.18%	+0.03
Triplet	1.55%	1.48%	-0.07
Contrastive	2.05%	2.10%	+0.05



Three aggregation strategies for the backbone's multimodal tokens:

- **Mean Pooling** — uniform average of all tokens
- **MQAP** — learned query, attention over the full sequence
- **Cross-Modal** — two independent queries (visual + text), learned fusion

Prompt conditioning: the backbone is frozen, but the *prompt* redirects cross-attention across all 36 layers — no weight is updated.

- $\mathcal{P}_0$: 4 tokens (minimal instruction)
- $\mathcal{P}_r$: 77 tokens (19× longer, layout-focused)

Quantitative ablation of pooler × prompt combinations follows next.

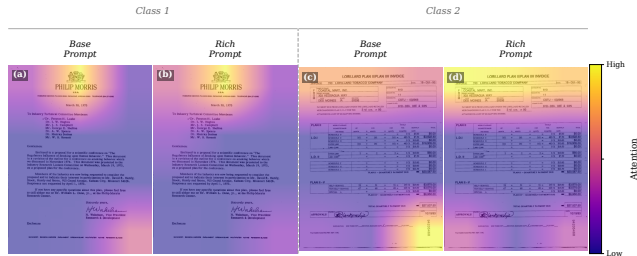


Figure: Mean text-to-visual attention at the backbone's output layer ($\mathcal{P}_0$ vs. $\mathcal{P}_r$, two example documents). $\mathcal{P}_0$ produces a diffuse distribution; $\mathcal{P}_r$ concentrates attention on layout-discriminative regions, with *no change to any backbone weight*.



Best loss (Sub-Center CosFace + *Teacher*)
carried forward to compare the three poolers
under $\mathcal{P}_0$ and $\mathcal{P}_r$.

- $\mathcal{P}_r$ improves **every** pooler, without exception
- **Mean Pool** + $\mathcal{P}_r$: best average EER ($0.63\% \pm 0.48$ pp)
- **Cross-Modal** + $\mathcal{P}_r$: $0.91\% \pm 1.10$ pp, wins on 4/5 individual splits
- MQAP + $\mathcal{P}_r$: $0.85\% \pm 0.55$ pp

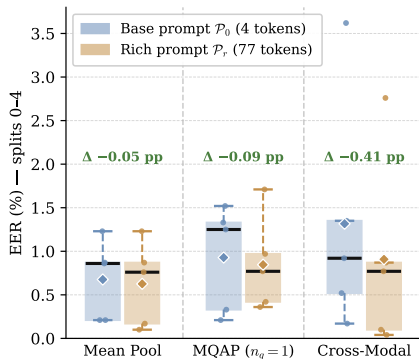


Figure: Cumulative EER (LA-CDIP, 5 splits) per pooler $\times$ prompt (Sub-Center CosFace). $\mathcal{P}_r$ improves every pooler.

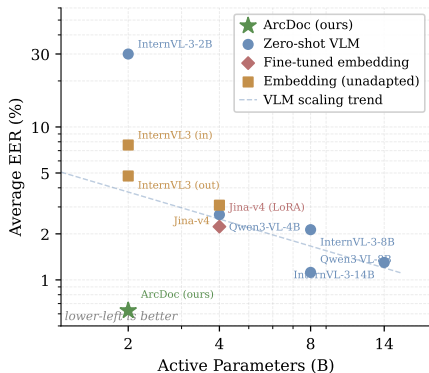


Figure: EER vs. active parameters (log-log). ArcDoc (star) operates 6× below the scaling trend of zero-shot VLMs.

Category	Method	Params	EER (%)
Ours	ArcDoc_{mean, P_r}	2B	0.63
	ArcDoc_{cross, P_r}	2B	0.91
Zero-Shot VLM	Qwen3-VL-8B	8B	1.12
	InternVL3-14B	14B	1.30
	InternVL3-8B	8B	2.13
	Qwen3-VL-4B	4B	2.67
	InternVL3-2B	2B	29.99
Embedding Fine-tuned	Jina-v4 (base)	4B	3.08
	Jina-v4 (LoRA $r=48$)	4B	2.23
Pixel	L2 / Cosine	—	31.2 / 32.3

InternVL3-2B zero-shot (same backbone, no training): 29.99% EER. **ArcDoc** (same backbone): 0.63% EER. A $\sim 48\times$ gap attributable exclusively to the ArcDoc angular margin head.



Configuration

- Loss: **Sub-Center CosFace** ($k=3$) + *Teacher*
- **1,748** original training images
- **6** variants per image
- **10,488** augmented images
- 120 training classes / 24 *novel*

Destination

Definitive model for publication on the **Hugging Face Hub**.

11 operations: Rotation, Perspective, Brightness/Contrast, Blur, Sharpening, Gaussian Noise, JPEG Q30–70, Resolution ↓, Crop/Padding, Yellowing, Salt & Pepper.

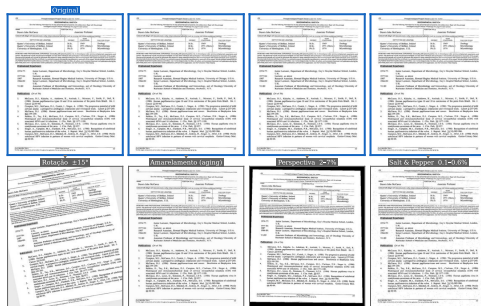
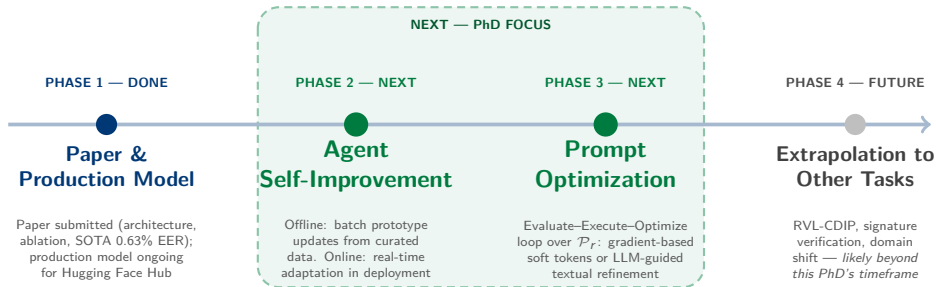


Figure: Four of the 11 operations (original → result). Training combines 6 at random per image.

Deployment-oriented pipeline, independent of the paper's evaluation protocol and splits.



Phase 1 (done): paper submitted, production model in progress. **Phases 2–3** (next, this PhD's focus): agent self-improvement, then prompt optimization. **Phase 4** (future): extrapolation to other document tasks — likely beyond this PhD's timeline.

Phase terminology: Liu et al., *Advances and Challenges in Foundation Agents*, arXiv:2504.01990 (2025) — Ch. 11 (*Online and Offline Agent Self-Improvement*), Sec. 9.2/Def. 9 (*Prompt Optimization*).



Thank You

Questions and discussion

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